

Budget Sets, Preferences & Choice : Problem Set Solutions

Problem 1

Let x_1 = pages in ECON
 x_2 = pages in SOC

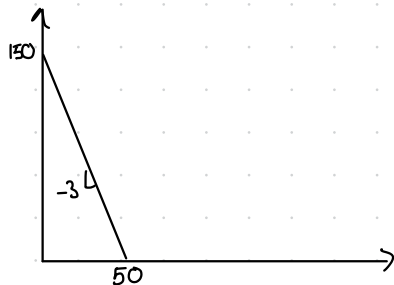
Two points: $(40, 30)$, $(30, 60) \rightarrow$ slope $m_1 = \frac{60-30}{30-40} = -\frac{30}{10} = -3$

Then the budget line is: $x_2 = 60 - 3(x_1 - 30) = 150 - 3x_1$

(a) 150 pages of SOC

(b) 50 pages of ECON

(c)



(d) For $x_1 \leq 25$, we are given two points $(10, 140)$, $(20, 130)$
 $\rightarrow$ slope $m_2 = \frac{130-140}{20-10} = -\frac{10}{10} = -1$

The first part of the budget line is:

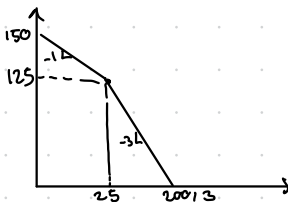
$$x_2 = 140 - (x_1 - 10) = 150 - x_1 \quad \text{for } x_1 \leq 25$$

There will be a kink at $x_1 = 25$, $x_2 = 150 - 25 = 125$

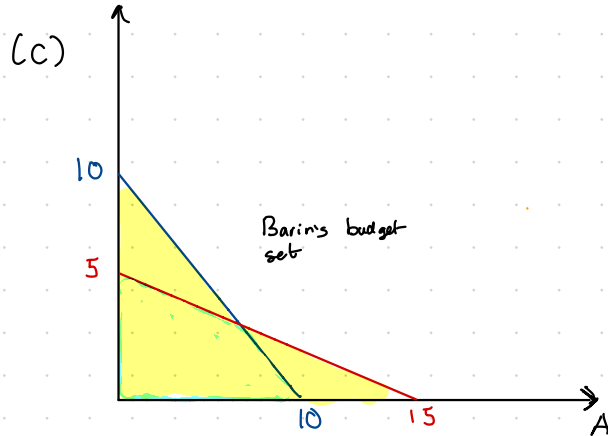
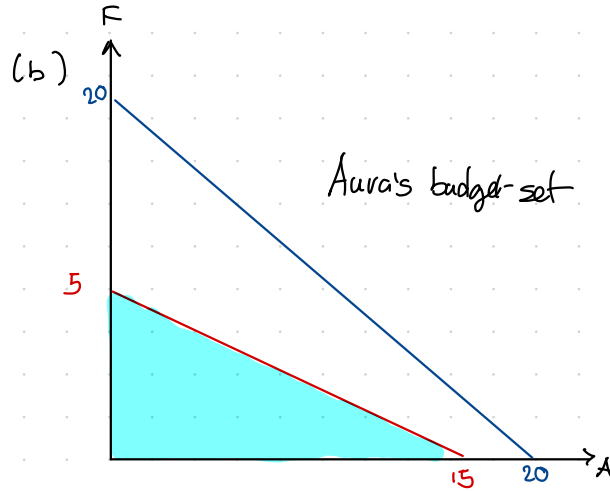
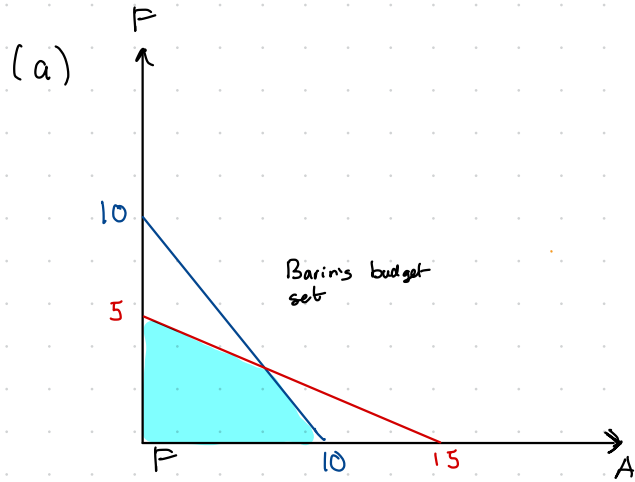
To obtain the second part we do:

$$x_2 = 125 - 3(x_1 - 25) = 200 - 3x_1 \quad \text{for } x_1 > 25$$

The x_1 intercept here is $x_1 = 200/3$



Problem 2:

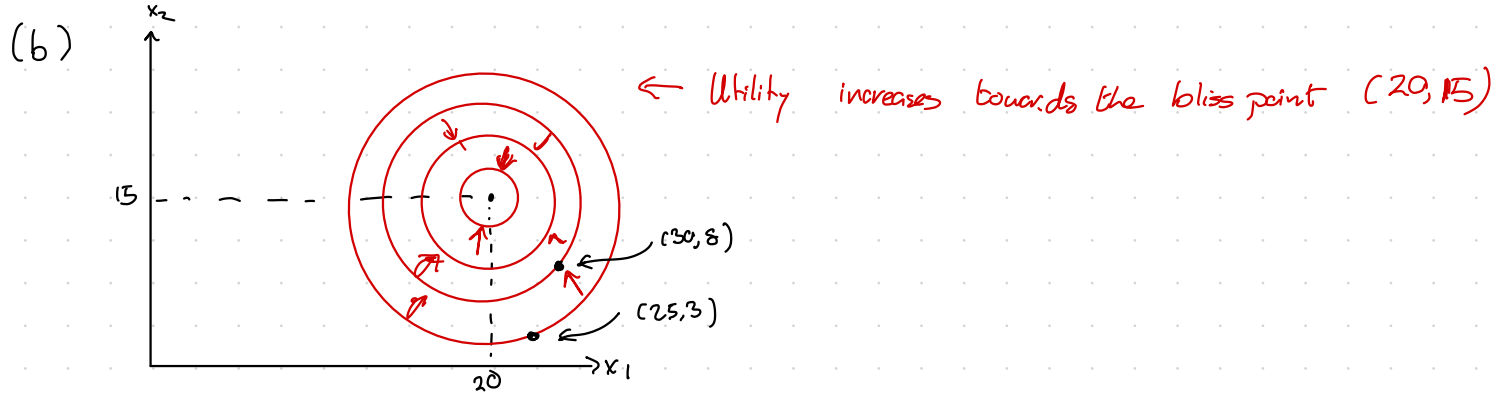


(d) No, examples:

- Time & money
- Money & Calorie budget
- Money & Alcohol budget

Problem 3

(a) A utility function representing her preferences is $U(x_1, x_2) = -[(x_1 - 20)^2 + (x_2 - 15)^2]$



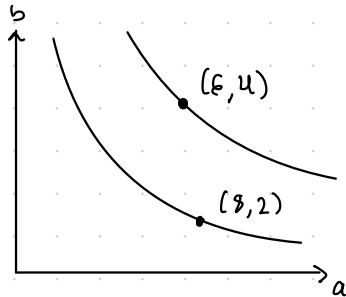
(c)
$$\left. \begin{aligned} U(25, 3) &= -169 \\ U(30, 8) &= -149 \end{aligned} \right\} \Rightarrow \text{Yes, she would prefer the combination } (30, 8)$$

(d)
$$MRS(25, 3) = \frac{25 - 20}{3 - 15} = -\frac{5}{12}$$

$$MRS(30, 8) = \frac{30 - 20}{8 - 15} = -\frac{10}{7}$$

Problem 4

(a) $u(8, 2) = \frac{16}{100} = 0.16$, $IC(8, 2): 0.16 = \frac{ab}{100} \Rightarrow 16 = ab \Rightarrow b = \frac{16}{a}$



$$u(6, 4) = \frac{6 \cdot 4}{100} = \frac{24}{100} = 0.24$$

$$IC(6, 4): 24 = ab \Rightarrow b = \frac{24}{a}$$

(b) $v(8, 2) = 1000 \cdot 8^2 \cdot 2^2$

$$v(6, 4) = 1000 \cdot 6^2 \cdot 4^2$$

$$IC(8, 2): 1000 \cdot 8^2 \cdot 2^2 = 1000 a^2 b^2 \\ \Rightarrow 16^2 = (ab)^2 \Rightarrow 16 = ab \Rightarrow b = \frac{16}{a}$$

$$IC(6, 4): 1000 \cdot 6^2 \cdot 4^2 = 1000 a^2 b^2 \\ \Rightarrow 24^2 = (ab)^2 \Rightarrow 24 = ab \Rightarrow b = \frac{24}{a}$$

$\therefore$ The same diagram as above.

(c) Yes for both

(d) They are the same preferences

(e) v is a monotonic transformation of u : $f(u) = 10000 u^2$

Problem 5

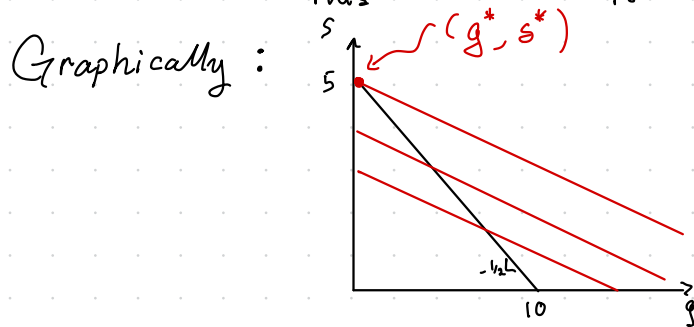
Let s represent the style index and g the gas index.

(a) The Budget Constraint is: $5,000s + 2,500g \leq 25,000$ or equivalently:

$$5s + 2.5g \leq 25. \quad \text{The Budget line is: } 5s + 2.5g = 25 \Rightarrow s = 5 - \frac{1}{2}g$$

(b) Zarina's preferences are of perfect substitutes. The indifference curves are straight lines with a slope of $MRS = 1/3$

Since $|MRS| = \frac{MU_g}{MU_s} = \frac{1}{3} < \frac{1}{2} = \frac{P_g}{P_s} \Rightarrow \underline{g^* = 0, s^* = 5}$

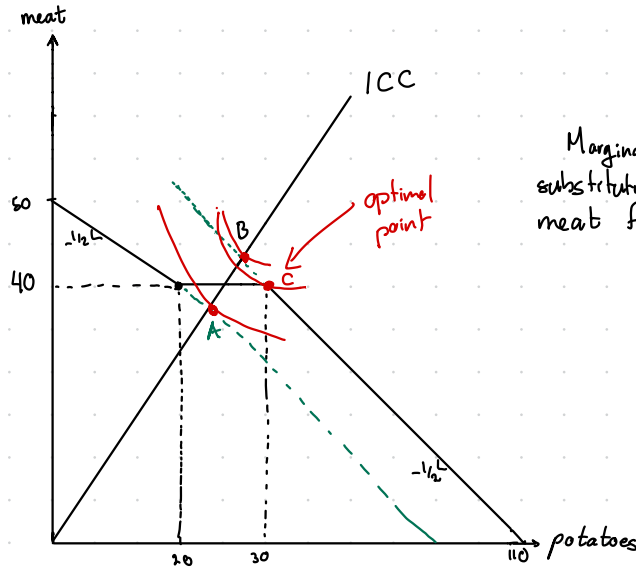


(c) $|MRS| = \frac{s}{4g}$. The optimality condition is $\frac{s}{4g} = \frac{1}{2}$
 $\Rightarrow s = 2g$. Plugging into the Budget line: $5(2g^*) + 2.5g^* = 25$
 $\Rightarrow 12.5g^* = 25 \Rightarrow \underline{g^* = 2, s^* = 4}$.

(d) $|MRS| = \frac{3s}{g}$. The optimality condition is $\frac{3s}{g} = \frac{1}{2} \Rightarrow 3s = g/2$
 $\Rightarrow g = 6s$. Plugging into the budget line: $5s^* + 2.5(6s^*) = 25$
 $5s^* + 15s^* = 25 \Rightarrow \underline{s^* = \frac{25}{20} = 1.25, g^* = 7.5}$.

Problem 6

(a)



The optimal point is C: in this point $|MRS| \neq \text{price ratio}$ but it is the highest indifference curve achievable in the budget set.

$$C = (30, 40)$$

To solve this problem we will need to use new tools.

Let's look at the income consumption curve

$$\text{Marginal rate of substitution of meat for potatoes} = \frac{M}{3P} = \frac{\text{price of potatoes}}{\text{price of meat}} = \frac{2}{4} = \frac{1}{2}$$

$$\Rightarrow M = \frac{3}{2} P$$

$$M = \frac{3}{2} (25) = \frac{75}{2}$$

Remark: the auxiliary budget lines marked as are just graphed to aid in understanding

↳ Notice that at point A, where

the IC would be tangent to the auxiliary Budget line, lies inside the budget set

Similarly, at point B, where the IC would be tangent to the auxiliary budget line, lies outside the budget set.

(b) The budget line becomes: $4M + 4P = 200$,

Thus the price ratio is now 1. The optimal consumption bundle is now obtained as follows:

$$\frac{M^*}{3P^*} = 1 \Rightarrow M^* = 3P^* \Rightarrow \text{Plug into the Budget Line: } 4(3P^*) + 4P^* = 200$$
$$\Rightarrow 16P^* = 200 \Rightarrow P^* = 12.5, M^* = 37.5$$

Problem 7

(a) $\frac{b}{2} = v$: equation for the "kink"

(b) Budget Line: $6b + 2v = 140$. Solving together with $v = b/2$ yields:

$$6b + 2\left(\frac{b}{2}\right) = 6b + b = 7b = 140 \Rightarrow b^* = 20, v^* = 10$$

(c) 10 martinis: 20 ounces of vodka + 10 ounces of vermouth make 30 ounces by volume; which means you can do $\frac{30}{3} = 10$ martini cocktails.

(d) 1 martini: 2 ounces of vodka (6) $\Rightarrow 6(2) + 2(1) = 12 + 2 = 14 //$
1 ounce of vermouth (v)